

Lecture - 11 [01 Sep 2026]

Policy Iteration and its Analysis

The policy iteration algorithm is used to find the optimal policy.

- Initialisation: $\phi: \mathcal{S} \rightarrow \mathcal{A}$ deterministic decision rule

$$\Gamma \leftarrow \mathcal{S}.$$

- Iterate: While Γ is non-empty:

- Policy evaluation

$$\underline{v}' \leftarrow (\mathbf{I} - \gamma P_{\phi})^{-1} r_{\phi}$$

$\underline{v}' \leftarrow$ value function of the policy $\pi = \{\phi, \phi, \dots\} \in \Pi^{\text{SD}}$

- Policy improvement

Set $\phi': \mathcal{S} \rightarrow \mathcal{A}$ as

$$\phi'(s) \in \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) v'(s') \right], \quad s \in \mathcal{S}.$$

Choose $\phi'(s) = \phi(s)$ if $\phi(s)$ is an element of $\arg \max$.

- $\Gamma \leftarrow \{s \in \mathcal{S} : \phi'(s) \neq \phi(s)\}.$

- $\phi \leftarrow \phi'$

- Upon termination, return ϕ and $\underline{v}'$

Remarks:

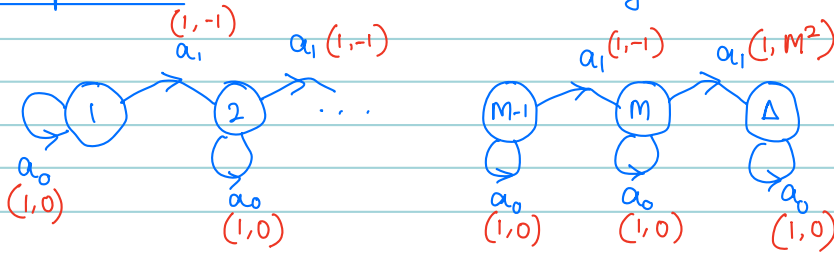
① The policy $\pi = \{\phi, \phi, \dots\}$, where ϕ is the decision rule returned by the policy iteration algorithm, is optimal.

② The value $\underline{v}'$ returned by the above algorithm is, in fact, $v_{\omega, \gamma}^*$, i.e;

$$\underline{v}' = v_{\omega, \gamma}^*$$

$$P(s'|s, a_0) = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & 0 & 1 \\ & & & 0 & 1 \end{bmatrix}$$

Example 5.14: Consider a time-homogeneous MDP given below.



$$\mathcal{S} = \{1, \dots, M, \Delta\}$$

$$A(s) = \{a_0, a_1\} \quad \forall s \in [M]$$

$$A(\Delta) = \{a_0\}$$

Consider the following implementation of policy iteration.

- Initialisation: $\phi(s) = a_0 \quad \forall s \in \mathcal{S}$

- For the above ϕ , we have $\pi_\phi = \underline{0}$, and therefore

$$\underline{v}' = (\mathbf{I} - \lambda P_\phi)^{-1} \pi_\phi = \underline{0}$$

$$\begin{aligned} \pi_\phi(s) &= \sum_a \phi(a|s) \pi(s, a) \\ &= \pi(s, \phi(s)) \end{aligned}$$

- Improvement:

$$\phi'(s) = \arg \max_a \left[\pi(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v'(s') \right]$$

$$\begin{aligned} &= 0, \quad \text{if } a = a_0, \\ &= -1, \quad \text{if } a = a_1, \quad s = 1, \dots, M-1 \\ &= M^2, \quad \text{if } s = M, \quad a = a_1 \end{aligned}$$

$$= \begin{cases} a_0, & s \neq M, \\ a_1, & s = M. \end{cases}$$

- $\Gamma \leftarrow \{s \in \mathcal{S} : \phi'(s) \neq \phi(s)\} = \{M\}$.

- $\phi \leftarrow \phi'$.

- For the updated ϕ , we have $\pi_\phi = \begin{bmatrix} 0, \dots, 0, M^2, 0 \end{bmatrix}^T$, and therefore

$$\forall s: \quad v'(s) - \lambda \sum_{s' \in \mathcal{S}} P(s'|s, \phi(s)) v'(s') = \pi(s, \phi(s))$$

$$\Leftrightarrow \quad v'(s) - \lambda v'(s) = 0, \quad s = 1, \dots, M-1$$

$$v'(M) - \lambda v'(\Delta) = M^2,$$

$$v'(\Delta) - \lambda v'(\Delta) = 0$$

$$\Leftrightarrow \quad \underline{v}' = \begin{bmatrix} 0, \dots, 0, M^2, 0 \end{bmatrix}$$

- The improvement step then gives

$$\phi'(s) = \begin{cases} a_0, & s=1, \dots, M-2, \Delta \\ a_1, & s=M-1, M. \end{cases}$$

- Proceeding this way, we eventually see that the algorithm terminates after $M+1$ iterations.

Convergence of Policy Iteration

The below result establishes that the sequence of value iterates generated by the policy iteration algorithm is non-decreasing. Moreover, a strict improvement leads to a strict increase in value.

Proposition 5.9

Let $\underline{v}' \in \mathbb{R}^{|S|}$ satisfy $(I - \lambda P_{\phi'}) \underline{v}' = r_{\phi}$. Let ϕ' be the rule

$$\phi'(s) = \arg \max_{a \in A} \left[r(s, a) + \lambda \sum_{s' \in S} P(s'|s, a) v'(s') \right],$$

with $\phi'(s) = \phi(s)$ whenever $\phi(s)$ is an element of $\arg \max$.

Then:

$$\begin{aligned} \textcircled{1} \quad r(s, \phi'(s)) + \lambda \sum_{s' \in S} P(s'|s, \phi'(s)) v'(s') \\ \stackrel{\textcircled{a}}{\geq} r(s, \phi(s)) + \lambda \sum_{s' \in S} P(s'|s, \phi(s)) v'(s') \end{aligned}$$

for all $s \in S$. In vector notation, $\textcircled{a}$ is same as

$$r_{\phi'} + \lambda P_{\phi'} \underline{v}' \geq r_{\phi} + \lambda P_{\phi} \underline{v}'.$$

$\textcircled{2}$ Suppose for some $s \in S$, the inequality $\textcircled{a}$ is strict, i.e., $\exists \delta > 0$ such that

$$r_{\phi'} + \lambda P_{\phi'} \underline{v}' = r_{\phi} + \lambda P_{\phi} \underline{v}' + \delta \underline{e}_s. \quad \text{unit vector with 1 in state } s$$

Then,

$$\underline{v}_{\infty, \lambda}^{\phi'} \geq \underline{v}' + \delta \underline{e}_s.$$

$$\phi \rightarrow \underline{v}' = (I - \lambda P_{\phi})^{-1} r_{\phi} \rightarrow \phi' \rightarrow \underline{v}_{\infty, \lambda}^{\phi'}$$

Proof of Proposition 5.9

Part ① follows from the definition of ϕ^1 as the argmax rule.

If the condition in part ② is satisfied for some $s \in \mathcal{S}$ and $\delta > 0$, then

$$\begin{aligned} r_{\phi^1} + \lambda P_{\phi^1} \underline{v}^1 &= r_{\phi} + \lambda P_{\phi} \underline{v}^1 + \delta \underline{e}_s \\ &= \underline{v}^1 + \delta \underline{e}_s \quad \left(\text{because } r_{\phi} + \lambda P_{\phi} \underline{v}^1 = \underline{v}^1 \right), \end{aligned}$$

from which it follows that

$$r_{\phi^1} = (\mathbf{I} - \lambda P_{\phi^1}) \underline{v}^1 + \delta \underline{e}_s.$$

Multiplying both sides by $(\mathbf{I} - \lambda P_{\phi^1})^{-1}$, we get

$$\begin{aligned} (\mathbf{I} - \lambda P_{\phi^1})^{-1} r_{\phi^1} &= \underline{v}^1 + \delta (\mathbf{I} - \lambda P_{\phi^1})^{-1} \underline{e}_s \\ \Leftrightarrow v_{\infty, \lambda}^{\phi^1} &= \underline{v}^1 + \delta (\mathbf{I} - \lambda P_{\phi^1})^{-1} \underline{e}_s \\ &= \underline{v}^1 + \delta \sum_{n=0}^{\infty} \lambda^n P_{\phi^1}^n \underline{e}_s \\ &\geq \underline{v}^1 + \delta \underline{e}_s \quad \left(\sum_{n=0}^{\infty} \lambda^n P_{\phi^1}^n \geq \mathbf{I} \right). \end{aligned}$$

Remark :

In the policy iteration algorithm, observe that $\underline{v}^1 = v_{\infty, \lambda}^{\phi}$.
According to Proposition 5.9,

$$v_{\infty, \lambda}^{\phi^1} \geq v_{\infty, \lambda}^{\phi},$$

with strict inequality if there is strict improvement in the policy improvement step.

Theorem 5.19 :

In an MDP with $\mathcal{S}, \mathcal{A}$ finite, the policy iteration algorithm converges monotonically to $v_{\infty, \lambda}^*$ in a finite number of iterations, and outputs the optimal policy and the optimal value function.

Proof: At each iteration epoch, either $\phi'(s) = \phi(s) \quad \forall s \in \mathcal{S}$, or there exists $s \in \mathcal{S}$ for which $\phi'(s)$ is strictly better than $\phi(s)$.

Because there are only finitely many deterministic decision rules possible when $\mathcal{S}, \mathcal{A}$ are finite, there must come an iteration when further improvement is not possible and $\phi' = \phi$.

At this point of no further improvement,

$$\underline{v}' = r_\phi + \lambda P_\phi \underline{v}' = r_{\phi'} + \lambda P_{\phi'} \underline{v}' = L \underline{v}',$$

thus implying that $\underline{v}' = \underline{v}_{\infty, \lambda}^*$ and $\phi' = \phi$ is optimal.

Policy Iteration and Newton's Method

In one-dimension, the Newton-Raphson method finds a solution x^* to the equation $f(x) = 0$ iteratively via (starting at an $x_0 \in \mathbb{R}$)

$$x_{k+1} = x_k - (f'(x_k))^{-1} f(x_k), \quad k=0, 1, 2, \dots \quad (\text{Newton-Raphson})$$

It is known that when f is convex, differentiable, and f' is Lipschitz, then Newton-Raphson method converges quadratically:

$$|x_{k+1} - x^*| \leq K |x_k - x^*|^2 \quad \forall k,$$

for some constant $K > 0$.

In higher dimension's, the Newton-Raphson method generalises as follows:

Suppose $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is given implicitly, and we wish to find $\underline{x}^*$ s.t.

$$F(\underline{x}^*) = \underline{0}.$$

We may find the solution $\underline{x}^*$ via (starting at an $\underline{x}_0 \in \mathbb{R}^n$)

$$\underline{x}_{k+1} = \underline{x}_k - (J(\underline{x}_k))^{-1} F(\underline{x}_k), \quad k=0, 1, \dots, \quad (\text{Newton-Raphson})$$

where $J(\underline{x}_k)$ denotes the Jacobian matrix of F evaluated at $\underline{x}_k$.

Recall that $B: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ defined as $B\underline{v} = L\underline{v} - \underline{v}$ satisfies the relation

$$F(\underline{v}) \geq F(\underline{u}) + (J(\underline{u})) (\underline{v} - \underline{u})$$

$$B_{\underline{v}} \geq B_{\underline{u}} + (\lambda P_{\phi_{\underline{u}}} - I) (\underline{v} - \underline{u})$$

Proposition 5.10 [Newton's method representation for Policy Iteration]

Let $\underline{v}'$ and $\underline{v}$ be the value iterates coming from the successive iterates ϕ' and ϕ of the policy iteration algorithm. Then,

$$\underline{v}'_{k+1} = \underline{v}_k - \underbrace{(\lambda P_{\phi_{\underline{v}}} - I)^{-1}}_{J(\underline{x}_k)^{-1}} \underbrace{B_{\underline{v}}}_{F(\underline{x}_k)}, \quad \phi \rightarrow \underline{v} = v_{\infty, \lambda}^{\phi} \rightarrow \phi' \rightarrow \underline{v}' = v_{0, \lambda}^{\phi'}$$

where $\phi_{\underline{v}} = \arg\max_{\phi} L_{\phi} \underline{v}$.

Proof of Proposition 5.10

Suppose that the policy iteration algorithm starts with ϕ and its value

$$\underline{v} = (I - \lambda P_{\phi})^{-1} r_{\phi} = v_{\infty, \lambda}^{\phi},$$

and proceeds to obtain $\phi_{\underline{v}} = \phi'$ in the policy improvement step.

Let $\underline{v}' = (I - \lambda P_{\phi_{\underline{v}}})^{-1} r_{\phi_{\underline{v}}}$ be the value function corresponding to ϕ' .

Then,

$$\begin{aligned} \underline{v}' &= (I - \lambda P_{\phi_{\underline{v}}})^{-1} r_{\phi_{\underline{v}}} + \underline{v} - \underline{v} \\ &= (I - \lambda P_{\phi_{\underline{v}}})^{-1} [r_{\phi_{\underline{v}}} + (\lambda P_{\phi_{\underline{v}}} - I) \underline{v}] + \underline{v} \\ &= (I - \lambda P_{\phi_{\underline{v}}})^{-1} [\underbrace{r_{\phi_{\underline{v}}} + \lambda P_{\phi_{\underline{v}}} \underline{v}}_{L_{\phi_{\underline{v}}} \underline{v} = L \underline{v}} - \underline{v}] + \underline{v} \\ &= (I - \lambda P_{\phi_{\underline{v}}})^{-1} [L \underline{v} - \underline{v}] + \underline{v} \\ &= \underline{v} - (\lambda P_{\phi_{\underline{v}}} - I)^{-1} B_{\underline{v}}. \end{aligned}$$

The following result provides a proof of convergence of policy iteration even when state/action spaces are not finite. It is based on the idea

that the value iterates coming from policy iteration are:

- ① Non-decreasing
- ② Bounded from below by the value iterates of value iteration.
- ③ Bounded from above by $v_{\infty, \lambda}^*$.

Thus, if value iterates of value iteration algorithm converge monotonically, then value iterates of policy iteration algorithm converge.

Note: One sufficient condition for iterates of value iteration to converge monotonically to $v_{\infty, \lambda}^*$ is that the initial iterate $\underline{v}_0$ satisfies

$$L\underline{v}_0 \geq \underline{v}_0.$$

$$\left[\underline{v}_0 = \frac{r^* \cdot 1}{1-\lambda}, \quad r^* := \min_{(s,a)} r(s,a) \right] \text{ satisfies } L\underline{v}_0 \geq \underline{v}_0.$$

Then,

$$\underline{v}_{k+1} = L^k L\underline{v}_0 \geq L^k \underline{v}_0 = \underline{v}_k \quad \forall k \geq 0.$$

Lemma 5.6:

For $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$, let $Z\underline{v} := \underline{v} - (\lambda P_{\phi_{\underline{v}}} - \mathbf{I})^{-1} B\underline{v}$.

Suppose that $\underline{v}$ satisfies $B\underline{v} \geq \underline{0}$. Then:
 $L\underline{v} - \underline{v} \geq \underline{0}$

$$\textcircled{1} \quad v_{\infty, \lambda}^* \geq Z\underline{v}.$$

$$\textcircled{2} \quad Z\underline{v} \geq L\underline{v}.$$

$$\textcircled{3} \quad Z\underline{v} \geq \underline{v}.$$

$$\textcircled{4} \quad B(Z\underline{v}) \geq \underline{0}.$$

Proof of Lemma 5.6:

① Observe that

$$Z\underline{v} = \underline{v} - (\lambda P_{\phi_{\underline{v}}} - \mathbf{I})^{-1} B\underline{v}$$

$$= (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} r_{\phi_{\underline{v}}}$$

$$= v_{\infty, \lambda}^{\phi_{\underline{v}}} \leq v_{\infty, \lambda}^*.$$

(reversing the steps in the proof of Proposition 5.10)

② This follows by noting that

$$Z_{\underline{v}} = \underline{v} + \left(\mathbf{I} - \lambda P_{\phi_{\underline{v}}} \right)^{-1} B_{\underline{v}}$$

$$\geq \underline{v} + B_{\underline{v}}$$

$$\left((\mathbf{I} - \lambda P_{\phi})^{-1} \underline{u} \geq \underline{u} \text{ whenever } \underline{u} \geq \underline{0} \right)$$

$$= L_{\underline{v}}.$$

③ If $B_{\underline{v}} \geq \underline{0}$, then $\left(\mathbf{I} - \lambda P_{\phi_{\underline{v}}} \right)^{-1} B_{\underline{v}} \geq \underline{0}$, and hence

$$Z_{\underline{v}} = \underline{v} + \left(\mathbf{I} - \lambda P_{\phi_{\underline{v}}} \right)^{-1} B_{\underline{v}}$$

$$\geq \underline{v}.$$

④ We have

$$Z_{\underline{v}} - \underline{v} = \left(\mathbf{I} - \lambda P_{\phi_{\underline{v}}} \right)^{-1} B_{\underline{v}},$$

and using $B_{\underline{v}} \geq B_{\underline{u}} + (\lambda P_{\phi_{\underline{u}}} - \mathbf{I})(\underline{v} - \underline{u})$ with $\underline{v} \leftarrow Z_{\underline{v}}$ and $\underline{u} \leftarrow \underline{v}$,

$$B(Z_{\underline{v}}) \geq B_{\underline{v}} + (\lambda P_{\phi_{\underline{v}}} - \mathbf{I})(Z_{\underline{v}} - \underline{v})$$

$$= B_{\underline{v}} + (\lambda P_{\phi_{\underline{v}}} - \mathbf{I}) \left(\mathbf{I} - \lambda P_{\phi_{\underline{v}}} \right)^{-1} B_{\underline{v}}$$

$$= B_{\underline{v}} - B_{\underline{v}}$$

$$= \underline{0}.$$

Theorem 5.33: Suppose $\underline{v}_0 \in \mathbb{R}^{|\mathcal{S}|}$ is such that $B_{\underline{v}_0} \geq \underline{0}$. Suppose further that for every $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$, there exists $\phi_{\underline{v}} : \mathcal{S} \rightarrow \mathcal{A}$ satisfying

$$\phi_{\underline{v}}(s) = \arg \max_{\phi} \left[r_{\phi} + \lambda P_{\phi} \underline{v} \right]$$

Then, the sequence of iterates $\{\underline{v}_m\}_{m \geq 0}$ of policy iteration converge monotonically to $\underline{v}_{\infty, \lambda}^*$.

The proof of Theorem 5.33 is a special case of a more general proof of convergence of modified policy iteration that will be covered in the next lecture.

Remark: By starting policy iteration with ϕ_0 and setting

$$\underline{v}_0 = (I - \lambda P_{\phi_0})^{-1} r_{\phi_0},$$

we see that $L_{\underline{v}_0} \geq \underline{v}_0$ is automatically satisfied:

$$\begin{aligned} L_{\underline{v}_0} &= \sup_{\phi} L_{\phi} \underline{v}_0 \\ &\geq L_{\phi_0} \underline{v}_0 = \underline{v}_0. \end{aligned}$$

Alternatively, policy improvement step can first be kickstarted as

$$\phi'(s) = \arg \max_a \left[r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s,a) v(s') \right],$$

where $\underline{v}$ satisfies $L_{\underline{v}} \geq \underline{v}$.

Policy Iteration vs Value Iteration

Value Iteration

- Performs

$$\underline{v}' \leftarrow L_{\underline{v}}$$

$$v'(s) = \arg \max_a \left[r(s,a) + \sum_{s' \in \mathcal{S}} P(s'|s,a) v(s') \right].$$

- Computationally efficient

- Returns an ϵ -optimal value $v_{\omega, \lambda, \epsilon}$:

$$\|v_{\omega, \lambda, \epsilon} - v_{\omega, \lambda}^*\|_{\infty} < \epsilon.$$

and ϵ -optimal policy ϕ_{ϵ} :

$$\|v_{\omega, \lambda}^{\phi_{\epsilon}} - v_{\omega, \lambda}^*\|_{\infty} < \epsilon$$

Policy Iteration

- Performs

$$\underline{v}' \leftarrow \underline{v} + (I - \lambda P_{\phi_{\underline{v}}})^{-1} B_{\underline{v}}$$

$$= L_{\underline{v}} + \lambda P_{\phi_{\underline{v}}} B_{\underline{v}} + \lambda^2 P_{\phi_{\underline{v}}}^2 B_{\underline{v}} + \dots$$

- Policy evaluation step

$$(I - \lambda P_{\phi})^{-1} r_{\phi}$$

is expensive; $O(|\mathcal{S}|^3)$ operations.

- Returns optimal policy and optimal value function

Remark: Many a time, ϕ_ε returned by value iteration is also optimal.